

STANDARD OPERATING PRACTICE

Running TLC the right way

Every rule, with the reason it exists — so plates become measurements

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WHY THIS DOCUMENT EXISTS

Reviewing our own plates (MEHQ series, Reaction_test1 screen) showed that most of them cannot yield an R_f value or a valid intensity comparison — not because the chemistry failed, but because of how the plate was run or photographed: no solvent-front line, uneven loading between lanes, reference lanes too faint to compare against, and photos taken at varying angles and formats. Each of those is a piece of information permanently lost. This standard closes those gaps.

THE ONE PRINCIPLE BEHIND EVERY RULE

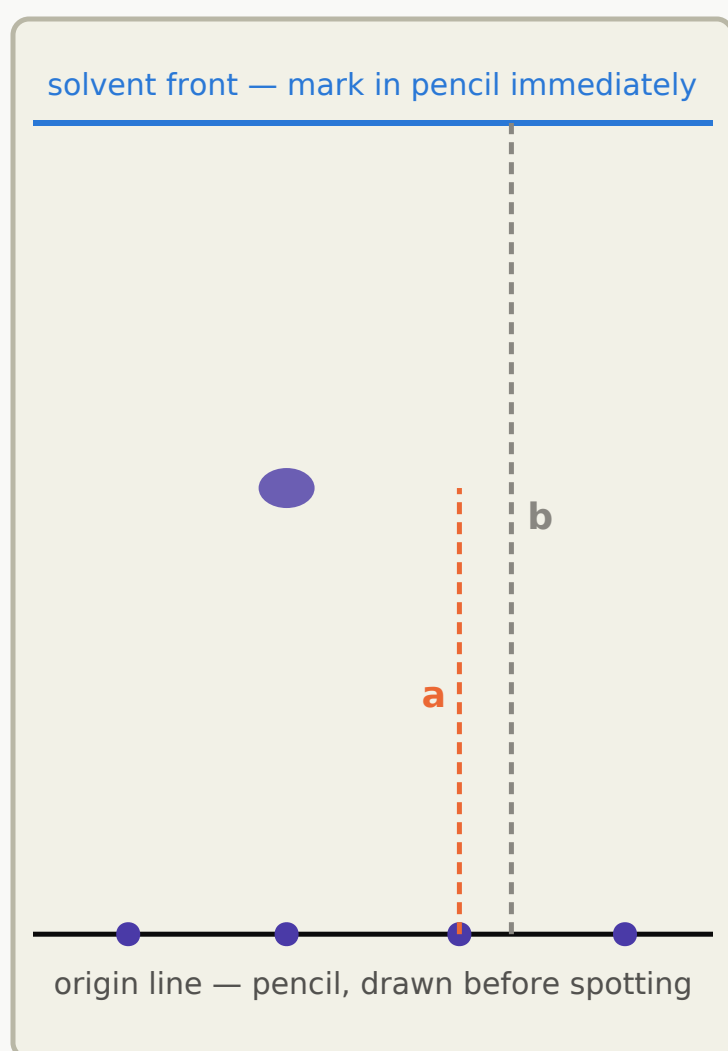
A TLC plate is a measurement, not a picture. Anything that is not recorded (a line, a label, a loading amount) and anything that is not kept constant (spot size, photo angle, lighting) becomes noise that no software can remove afterwards.

WHAT THIS STANDARD IS GROUNDED IN

Published, validated practice: smartphone JPEG photos are sufficient for quantitative TLC — the TLCyzer study (Nature Scientific Reports, 2022) reached ~100% mean accuracy and ~3% repeatability from ordinary phone images, and phone-camera TLC has been validated for drug-authenticity screening (J. Pharm. Biomed. Anal., 2016). Standardised sample application is the known precision bottleneck in planar chromatography (CAMAG / HPTLC practice). No RAW/DNG files, special cameras, or instruments are required — only consistency.

Draw both lines, always

The origin line and the solvent-front line are what turn spot positions into numbers



$$R_f = a / b$$

distance the spot travelled (a) ÷ distance the solvent travelled (b), both measured from the origin line.

1 Draw the origin line before spotting

Light pencil line ~1.5 cm from the bottom edge. Spot every sample exactly on it. Pencil only — ink runs with the solvent.

WHY: R_f 's numerator and denominator both start here. If spots sit at different heights, every distance is measured from a different zero and lanes stop being comparable.

2 Mark the solvent front immediately

The moment the plate leaves the chamber, pencil-mark where the solvent reached — within seconds, before it evaporates.

WHY: The front dries without a trace within minutes. Without this line R_f cannot be computed — not by eye, not by software. All our MEHQ plates miss it.

CAN SOFTWARE DRAW THE LINES VIRTUALLY? — THE HONEST ANSWER

The origin line: mostly yes. The spotting dots usually remain visible, and a line fitted through them recovers the origin. But it fails exactly when plates are messy — when streaks or halos swallow the dots (our P32 plates) the estimate degrades, and a fitted line always carries more error than a drawn one. So: software can rescue an origin, but the pencil line stays the rule.

The solvent front: no. Once the solvent evaporates there is nothing in the image to find — no edge, no tint, no texture. A front that was not pencil-marked is gone permanently, and with it every R_f on that plate. This is the one mistake software can never repair, which is why rule 2 is absolute.

Equal spots, equal footing

Intensity comparisons are only valid when every lane is loaded the same way

3 Same capillary, same number of touches, every lane

Use one capillary size for the whole plate. Load each lane with the same number of touches (e.g. 2×), letting the spot dry between touches. Keep spots $\leq 3\text{--}4$ mm.

WHY: Spot darkness is how amount is read. If one lane got one light touch and another got three heavy ones, 'darker' no longer means 'more compound' — it means 'more loaded'. Comparison dies.

4 Load the reference lanes to comparable strength

S and Std must be visible at roughly the same intensity scale as the reaction lanes — dilute or concentrate the references to match.

WHY: A reference that is barely visible cannot anchor any comparison. On our MEHQ-P33 plate the Std spot is so faint that 'how much product formed vs standard' is unanswerable.

5 Space the lanes; keep margins

≥ 8 mm between spots, ≥ 10 mm from each plate edge. Pencil-label every lane under the origin line.

WHY: Close spots bleed into each other as they develop — on our E-30min plate the S spot overflows into the A₂ lane, contaminating its lane read. Edge spots run crooked (edge effect).

6 Don't overload

If a lane streaks from origin to mid-plate, dilute the sample and respot. A good spot stays a compact ellipse after development.

WHY: Overloading smears compounds into long tails that bury neighbouring spots and destroy position and intensity information. Our screening plates (T-30, E-30) show full-length streaks in every reaction lane.

Run and label the plate the same way, every time

Rf is only reproducible when the run conditions are held constant

7 Saturated, lidded chamber; solvent below the origin

Line the chamber with filter paper, let it saturate ~5–10 min, keep the lid on. Solvent depth must stay below the origin line.

WHY: An unsaturated chamber makes solvent evaporate off the plate mid-run, bending Rf values. Solvent above the origin dissolves your spots into the pool before they ever run.

8 Stop the run at a fixed point

Remove the plate when the front is ~1 cm from the top — the same stopping rule every run. Then mark the front (rule 2) and let the plate dry flat.

WHY: Letting the front hit the top edge destroys the denominator of Rf (you no longer know how far the solvent 'really' went) and compresses spots into the top margin.

9 Write the metadata on the plate, in one format

Top edge, above the front line: reaction ID, timepoint, mobile phase — e.g. 'MEHQ-P34 · 60 min · 30% EA-Hex'. Lane labels under the origin.

WHY: The plate is the record. Our Reaction_test1 plates do this well (T·30 min etc.) — the photo then carries its own identity and can never be mismatched to a reaction.

10 Keep the TLC solvent system fit for the question

Aim for the compound of interest around Rf 0.3–0.5, and for visible separation between S and Std. If the two references co-run, change the ratio before trusting any plates.

WHY: On all three 30-min screening plates, S and Std sit at nearly the same height — 'SM consumed' and 'product formed' become almost indistinguishable. A weaker eluent separates the question itself.

A normal smartphone JPEG is enough — taken consistently

Published work quantifies plates from phone JPEGs; RAW/DNG adds size, not information

11 Phone camera, JPEG, UV box on — nothing fancier

Ordinary phone JPEG in the UV-254 cabinet. No RAW/DNG (our 45 MB DNG files add nothing for spot reading), no flash, no zoom.

WHY: Smartphone JPEGs are validated for quantitative TLC (~3% error in the TLCyzer study). What matters is not the format but the consistency of the shot.

12 Straight-on, plate filling most of the frame

Hold the phone parallel to the plate (90° view), plate filling ~60–80% of the frame, origin at the bottom. Tap to focus on the plate.

WHY: Tilt turns rectangles into trapezoids and shifts every measured distance. A plate photographed from far away (our IMG_0450: the plate is a stamp in the corner of the frame) has too few pixels per spot to measure anything.

13 Whole plate in frame: both lines, all labels, header

The photo must include the origin line, the marked front line, every lane label and the header text. No cropping at the edges.

WHY: Every one of those items is data the system reads. A cropped front line = no R_f. A cropped label = an unidentifiable lane.

14 One photo per plate; name the file like the plate

One clean shot, filename = plate header: MEHQ-P34_T_30min.jpg. Re-shoot if there's glare, blur, or a shadow across the plate.

WHY: Duplicate and unnamed files (IMG_0454.DNG...) force someone to re-read handwriting to know what a photo is. The filename should answer it instantly.

OPTIONAL UPGRADES (NOT REQUIRED)

A phone tripod/stand in front of the UV box fixes distance and angle for free. Locking exposure (long-press on iPhone) keeps brightness comparable across a series. Both help; neither is a precondition.

Where each rule comes from — our actual images

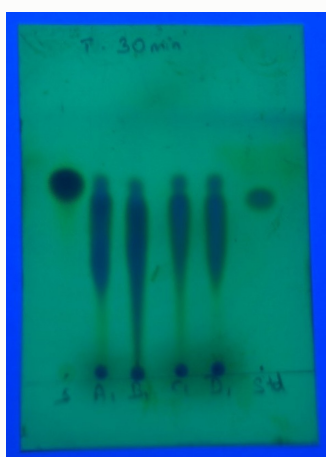
Every rule in this standard traces back to a real plate of ours



MEHQ-P33 — no solvent-front line, faint Std

No pencil front mark anywhere on the plate, so no R_f can ever be computed from this photo — only relative positions. And the Std lane is so lightly loaded that product intensity cannot be judged against it.

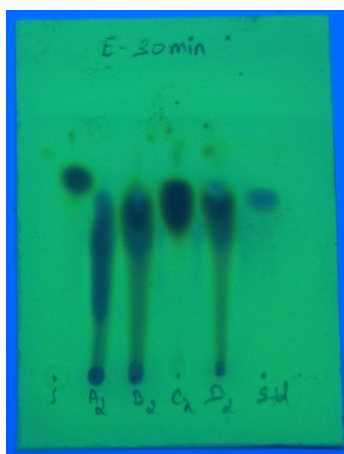
→ Rules 2, 4



T-30 min — lines done right, loading done wrong

Both pencil lines are present and the header is exemplary (rule 9 done well). But every reaction lane streaks from origin to mid-plate — heavy loading smears the lanes and buries any separate spots inside the tails.

→ Rules 3, 6



E-30 min — spot spacing too tight

The S reference spot bleeds sideways into the A₂ lane. Whatever the A₂ lane now shows near that height is partly contamination from its neighbour, not chemistry.

→ Rule 5



IMG_0450 — plate too small in the frame

Shot from across the cabinet: the plate occupies a tiny fraction of the image, leaving too few pixels per spot for any measurement. Also stored as unnamed HEIC/DNG files that duplicate each other.

→ Rules 11, 12, 14

The whole standard on one page

Pin this next to the TLC bench

Before spotting

- ☐ pencil origin line ~1.5 cm from bottom
- ☐ lane labels written under the line
- ☐ header: ID · timepoint · mobile phase
- ☐ references diluted to comparable strength

Spotting

- ☐ same capillary, same touches per lane
- ☐ spots on the line, ≤ 3–4 mm
- ☐ ≥ 8 mm apart, ≥ 10 mm from edges
- ☐ dry between touches

Development

- ☐ saturated chamber, lid on
- ☐ solvent level below origin line
- ☐ stop ~1 cm from the top
- ☐ mark the front IMMEDIATELY

Photo

- ☐ UV box on, phone JPEG, no flash
- ☐ straight-on, plate fills the frame
- ☐ both lines + all labels visible
- ☐ filename = plate header

WHAT SOFTWARE CAN AND CANNOT FIX AFTERWARDS

Problem	Fixable?	Detail
Rotated / upside-down photo	fixable	auto-rotation is trivial
Mild tilt or perspective	fixable	corrected from the plate's rectangle
Missing origin line	partly	fitted through spotting dots — degrades on messy plates
Missing solvent-front line	never	nothing remains in the image to find; R _f is lost
Under-loaded reference lane	never	information was never captured on the plate
Overloading streaks	never	spot positions and amounts destroyed by the smear
Glare / blur / low resolution	rarely	re-shoot instead — pixels can't be invented

The pattern in this table is the message of the whole document: software forgives photography mistakes, partially forgives missing origins, and never forgives a missing front line, uneven loading, or overloading. Those three must be right on the bench.